Group-Combined P-values with Applications to Genetic Association Studies

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Outline

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Introduction

To detect the association between a genomic region with hundreds of thousands of single nucleotide polymorphisms (SNPs) and human complex diseases in large scale genetic studies.

$$logit\ P(Y = 1|X) = X_1\beta_1 + X_2\beta_2 + \cdots + X_m\beta_m \tag{1}$$

To test

$$H_0 : \beta_1 = \cdots = \beta_m = 0 \ \text{ vs } \ H_1 : \exists\beta_i \neq 0 (i = 1, \ldots, m) \tag{2}$$

The logistic regression method: work inappropriately for a high-dimensional problem

P-value combination procedures : a useful strategy constructed from single-marker analysis

Previous methods

- Fisher's combined test (Fisher,1932)
- Truncated product method (Zaykin,2002)
- Tail strength method (Taylor & Tibshirani,2006)
- Truncated tail strength method (Jiang,2011)
- Rank truncated product method (Dudbridge & Koeleman,2003)
- Adaptive rank truncated product method (Yu,2009)
- Sequential test(Chen,2013)

Previous methods

Fisher(1932) proposed to use the product of all p-values in test statistic

$$FCT = -2\sum_{i=1}^{m} \ln p_i = -2\ln(\prod_{i=1}^{m} p_i) \tag{3}$$

If the combined m p-values are independent, the $FCT|H_0 \sim \chi^2_{2m}$.
If m p-values are correlated, the re-sample methods are recommended to get the empirical distribution of FCT.
The FCT has been widely applied in genetic association studies (Hess,2007;Zheng,2012;Li,2014). The FCT might lose power when the number of degrees of freedom is large and most of the SNPs are not significant.

Previous methods

To overcome the drawbacks, combining the truncated p-values were recommended.
Zaykin(2002) proposed a truncated product method (TPM) by using the product of p-values less than some specified threshold.

$$TPM(p_1, p_2, \ldots, p_m) = \prod_{i=1}^{m} p_i^{I(p_i \leq \tau)} \tag{4}$$

By choosing an proper cut-point such as 0.05, the TPM will increase the power.

Previous methods

Dudbridge and Koeleman(2003) proposed forming the product of the K most significant p-values.

$$RTP(p_1, p_2, \ldots, p_K) = \prod_{i=1}^{K} p_{(i)} \tag{5}$$

Its power is very sensitive to K and an inappropriate K result in combing some non-significant p-values and thus it loses power.

Previous methods

Yu et al. (2009) suggested using some candidate truncation points $K_1, \ldots, K_J$ and proposed an adaptive rank truncated product method (ARTP)

$$ARTP = min_{j=1,\ldots,J} q(K_j) \tag{6}$$

where $q(K_j)$ is the estimated p-value for $RTP(K_j), j = 1, \ldots, J$.

Previous methods

Taylor and Tibshirani(2006) defined the tail strength(TS) method as

$$TS(p_1, p_2, \ldots, p_m) = \frac{1}{m}\sum_{i=1}^{m}(1 - p_{(i)}\frac{m+1}{i}) \tag{7}$$

Jiang et al.(2011) proposed a truncated tail strength statistic (TTS), which was proved to have higher power than the tail strength method.

$$TTS(p_1, p_2, \ldots, p_m) = \frac{1}{m}\sum_{i=1}^{m} I(p_{(i)} \leq \tau)(1 - p_{(i)}\frac{m+1}{i}) \tag{8}$$

Our method

- The truncated combined p-value procedures (TPM,RTP,ARTP,TTS): P-values are divided into two groups and the small p-values are used to construct the test statistics.
- Our method: Dividing the p-values into three or three more groups and we construct a test statistic in each group and then combine them to form an omnibus test.

Our method

A nature extension is to divide the p-values in k groups and combine the p-values at the group level and form a group-combined test statistics.
Let $c_1, c_2, \cdots, c_k$ with $0 < c_1 < c_2 < \cdots < c_k < 1$ be k thresholds. Then the group-combined p-values test statistic (GCP) is constructed as following

$$\text{GCP} = \prod_{i=1}^{k} \left[1 - F_i \left(\sum_{j=1}^{m} -2 \ln p_{(j)} I_{\{c_{i-1} < p_{(j)} \leq c_i\}} \right) \right],$$

where $c_0 = 0$, and $F_i(\cdot)$ is the cumulative distribution of $\sum_{i=1}^{m} -2 \ln p_{(i)} I_{\{c_{i-1} < p_{(i)} \leq c_i\}}$ under null hypothesis for $i = 1, 2, \cdots, k$.

Our method

Algorithm:

Step 1:

- For s from 1 to B, permutate the original data under the null hypothesis and calculate m p-values, denote them by $p_1^{(s)}, \ldots, p_m^{(s)}$.
- Using them to calculate the empirical cumulative distribution $F_i(\cdot)$ for $i = 1, 2, \ldots, k$ and the corresponding GCP, denote it by GCP_s.

Step 2:
Calculate GCP using the observed data, denote it by GCP_0.
Step 3:

The p-value of GCP is $\dfrac{\#\left\{\mathrm{GCP}_s > \mathrm{GCP}_0 \middle| j=1,2,\cdots,B\right\}}{B}$.

Simulations

- We conduct simulation studies to explore the performances of the proposed GCP by comparing with FCT, TPM, TTS and ARTP.
- For TPM, we choose $\tau = 0.05$ as the cut point. For ARTP, the candidate points $K_j = \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$ recommended by Chen et al. (2013) are used.
- We consider $m = 100, 200$ and among m SNPs, at most $L\ (= m^{1-\gamma})$ SNPs are allowed to be associated with the phenotype (Mukherjee et al,2015), where γ is used to determine whether the signals are dense or sparse ($\gamma \leq \frac{1}{2}$ corresponds to a dense SNPs region and $\gamma \geq \frac{1}{2}$ corresponds to a sparse regime). We set $\gamma = 0.6$.

Group Number and Threshold

Independent case

Generate m independent random observations, and $m - L$ p-values are from $U(0,1)$ distribution under null hypothesis, L p-values from *Beta* distribution with fixed parameter $a = 0.3, b = 30$.

Dependent case

Generate m correlated p-values from a copula, and the structures of variance-covariance matrixs we use are defined by

$$\rho_1 = \begin{pmatrix} 1 & \dots & & \\ \dots & 1 & \rho & \\ & \rho & 1 & \dots \\ & & \dots & 1 \end{pmatrix}, \ \rho_2 = \begin{pmatrix} 1 & \dots & & \\ \dots & 1 & \rho^{|i-j|} & \\ & \rho^{|i-j|} & 1 & \dots \\ & & \dots & 1 \end{pmatrix} \tag{9}$$

$m - L$ p-values under null hypothesis have $U(0,1)$ marginal distribution, and under alternative hypothese, we transform $p_1, \dots, p_L$ to a beta distribution.

Group Number and Threshold

Table 1: Performance comparisons of GCP in several cases when L of $m = 100$ hypotheses are alternative

Methods (Independent case)	L				
	0	1	2	3	4
GCP($\tau_1 = 0.01, \tau_2 = 0.1$)	0.049	0.197	0.429	0.670	0.860
GCP($\tau_1 = 0.001, \tau_2 = 0.05$)	0.043	0.315	0.574	0.762	0.893
GCP($\tau_1 = 0.01, \tau_2 = 0.05$)	0.051	0.191	0.421	0.691	0.87
GCP($\tau_1 = 0.001, \tau_2 = 0.01, \tau_3 = 0.1$)	0.048	0.276	0.559	0.774	0.895
GCP($\tau_1 = 0.001, \tau_2 = 0.01, \tau_3 = 0.05$)	0.045	0.304	0.568	0.777	0.904

Group Number and Threshold

Methods (Dependent case 1)	L				
	0	1	2	3	4
GCP($\tau_1 = 0.01, \tau_2 = 0.1$)	0.049	0.056	0.072	0.080	0.106
GCP($\tau_1 = 0.001, \tau_2 = 0.05$)	0.045	0.361	0.542	0.640	0.693
GCP($\tau_1 = 0.01, \tau_2 = 0.05$)	0.046	0.054	0.072	0.087	0.097
GCP($\tau_1 = 0.001, \tau_2 = 0.01, \tau_3 = 0.1$)	0.050	0.154	0.233	0.415	0.47
GCP($\tau_1 = 0.001, \tau_2 = 0.01, \tau_3 = 0.05$)	0.048	0.157	0.278	0.331	0.443

Group Number and Threshold

Methods (Dependent case 2)	L				
	0	1	2	3	4
GCP($\tau_1 = 0.01, \tau_2 = 0.1$)	0.048	0.102	0.242	0.417	0.617
GCP($\tau_1 = 0.001, \tau_2 = 0.05$)	0.050	0.250	0.489	0.620	0.731
GCP($\tau_1 = 0.01, \tau_2 = 0.05$)	0.048	0.111	0.267	0.450	0.631
GCP($\tau_1 = 0.001, \tau_2 = 0.01, \tau_3 = 0.1$)	0.053	0.198	0.415	0.595	0.712
GCP($\tau_1 = 0.001, \tau_2 = 0.01, \tau_3 = 0.05$)	0.051	0.199	0.422	0.613	0.734

Group Number and Threshold

From Table 1, obviously *Group Number* $= 2$ with the combination of $c_1 = 0.001$ and $c_2 = 0.05$ are powerful than other combiation. We recommend to using the *GCP* as

$$\mathrm{GCP} = \left[1 - F_1\left(\sum_{j=1}^{m} -2\ln p_{(j)} I_{\{0<p_{(j)}\leq 0.001\}}\right)\right] \times \left[1 - F_2\left(\sum_{j=1}^{m} -2\ln p_{(j)} I_{\{0.001<p_{(j)}\leq 0.05\}}\right)\right]$$

Empirical Type I Error Rates and Power

Scenario 1		Independent		
Scenario 2	$m = 100$	$\rho_1 \& \rho_2$	0.5/0.8	Gaussian-copula
Scenario 3	$m = 200$	$\rho_1 \& \rho_2$	0.5/0.8	Gaussian-copula
Scenario 4	$m = 100$	$\rho_1 \& \rho_2$	0.5/0.8	t-copula
Scenario 5	$m = 200$	$\rho_1 \& \rho_2$	0.5/0.8	t-copula

Type I Error Rates

Table: Type-I error rates of the tests when all m hypotheses are true at the significance level of 5%.

Methods (Scenario 1)	m	
	100	200
FCT	0.053	0.049
TPM	0.0482	0.0516
GCP	0.0485	0.0462
TTS	0.0489	0.0509
ARTP	0.0547	0.0518

Type I Error Rates

Methods (Scenario 2)	ρ_1		ρ_2	
	0.5	0.8	0.5	0.8
FCT	0.0482	0.0455	0.049	0.0503
TPM	0.0539	0.0465	0.0524	0.0591
GCP	0.0517	0.049	0.0537	0.045
TTS	0.0506	0.0448	0.0496	0.0541
ARTP	0.0502	0.0461	0.0459	0.054

Methods (Scenario 3)	ρ_1		ρ_2	
	0.5	0.8	0.5	0.8
FCT	0.05039	0.0524	0.0475	0.0416
TPM	0.053	0.0464	0.0483	0.0441
GCP	0.05	0.0505	0.0474	0.0497
TTS	0.0522	0.0514	0.0487	0.0439
ARTP	0.0565	0.0513	0.0524	0.047

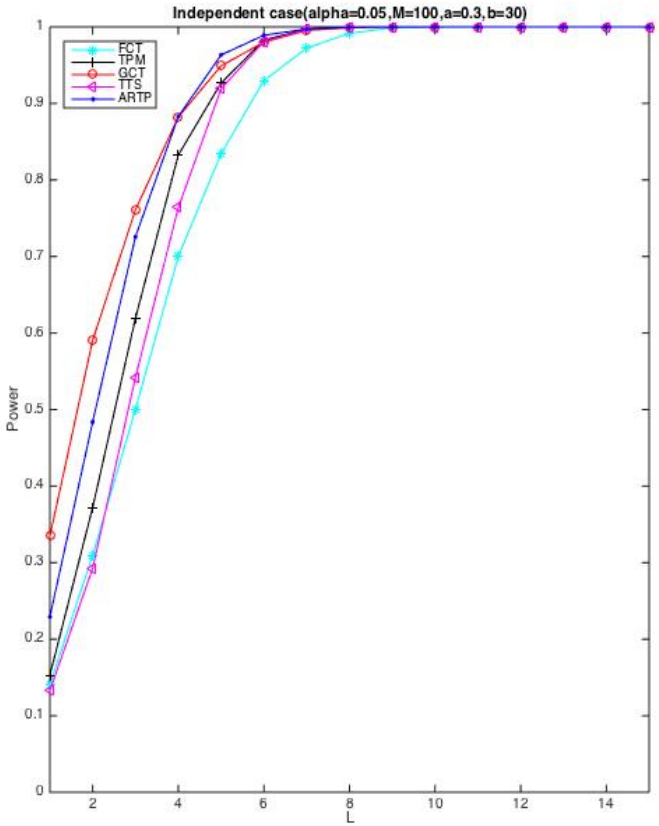
Type I Error Rates

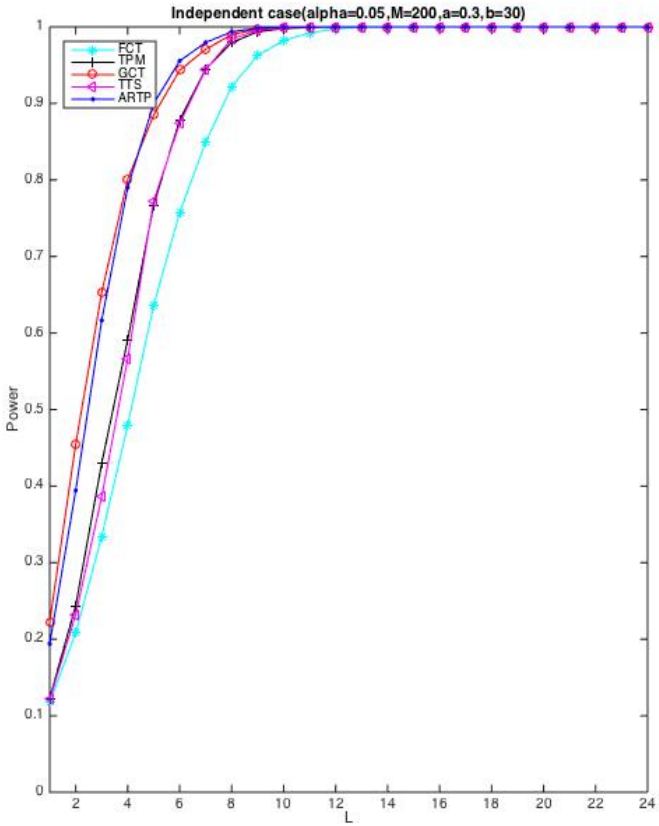
Methods (Scenario 4)	ρ_1		ρ_2	
	0.5	0.8	0.5	0.8
FCT	0.0459	0.0495	0.0457	0.052
TPM	0.0503	0.0502	0.0483	0.0485
GCP	0.0468	0.0503	0.0487	0.0488
TTS	0.0508	0.0498	0.0455	0.05
ARTP	0.0444	0.0492	0.0449	0.0472

Methods (Scenario 5)	ρ_1		ρ_2	
	0.5	0.8	0.5	0.8
FCT	0.0523	0.0517	0.0526	0.0545
TPM	0.048	0.0519	0.0487	0.05
GCP	0.0539	0.0497	0.0502	0.0527
TTS	0.0457	0.0478	0.0483	0.0516
ARTP	0.0498	0.0459	0.0528	0.05

Power Comparison

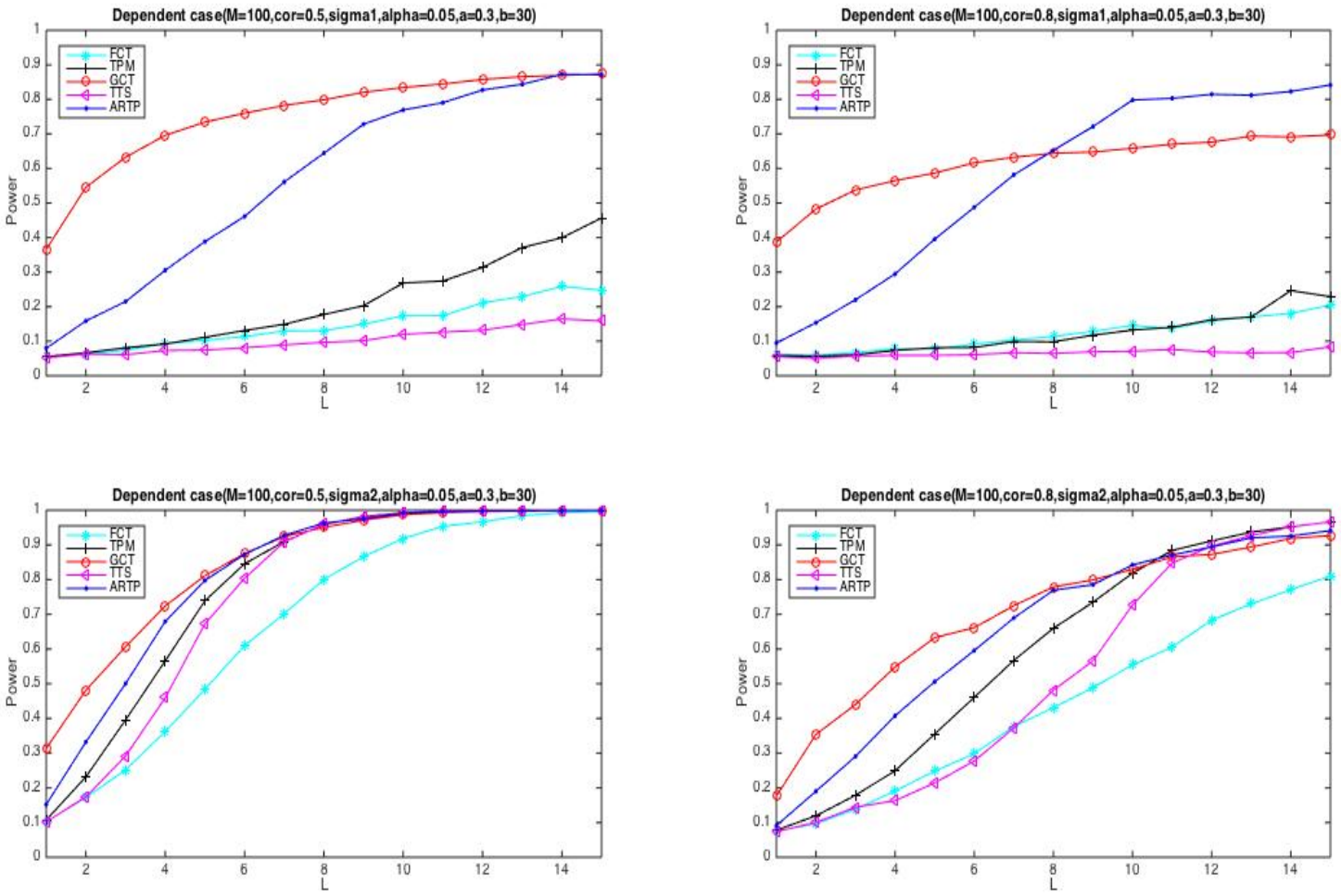
Figure 1: Power comparison of Scenario 1





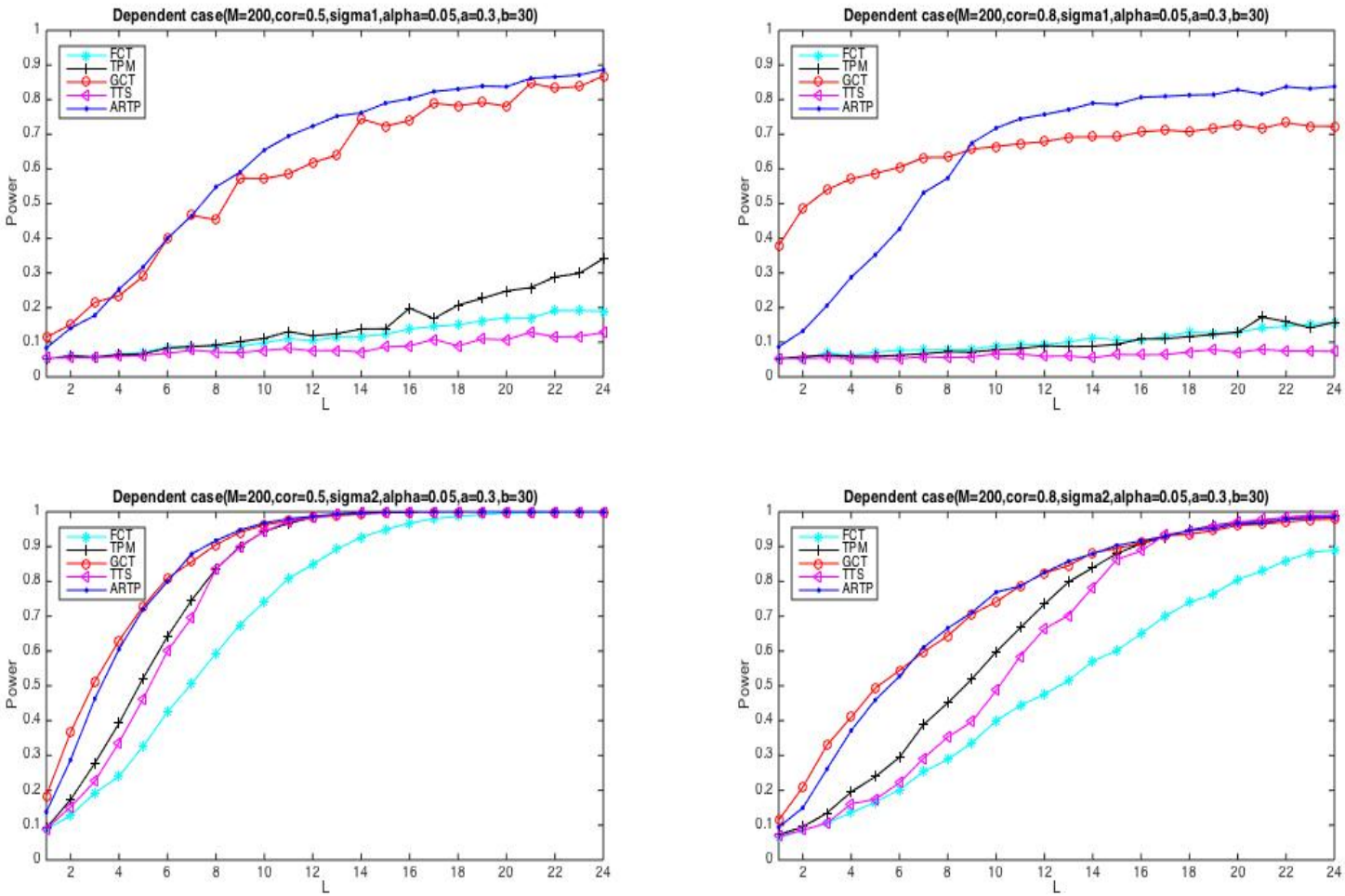
Power Comparison

Figure 2: Power comparison of Scenario 2



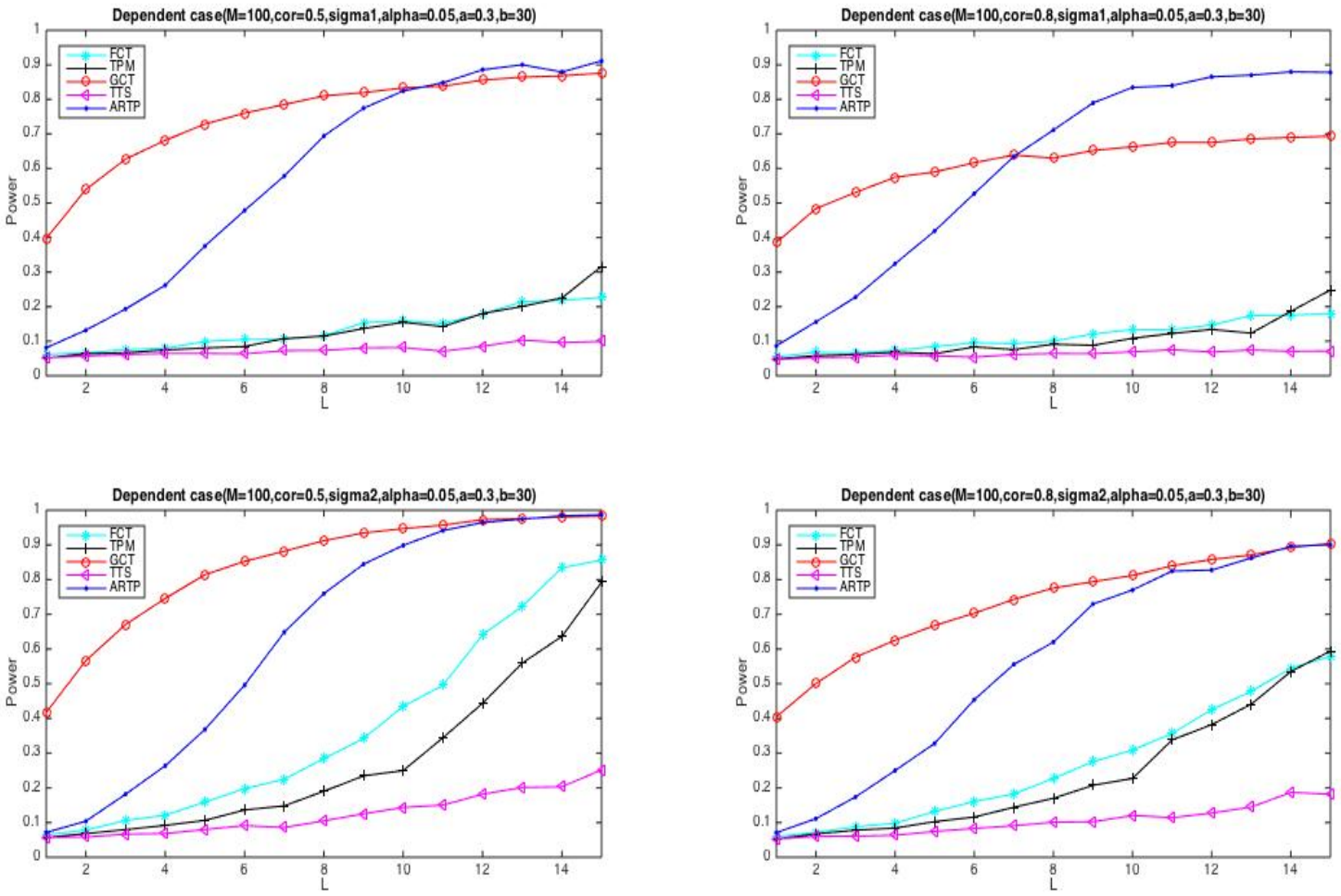
Power Comparison

Figure 3: Power comparison of Scenario 3



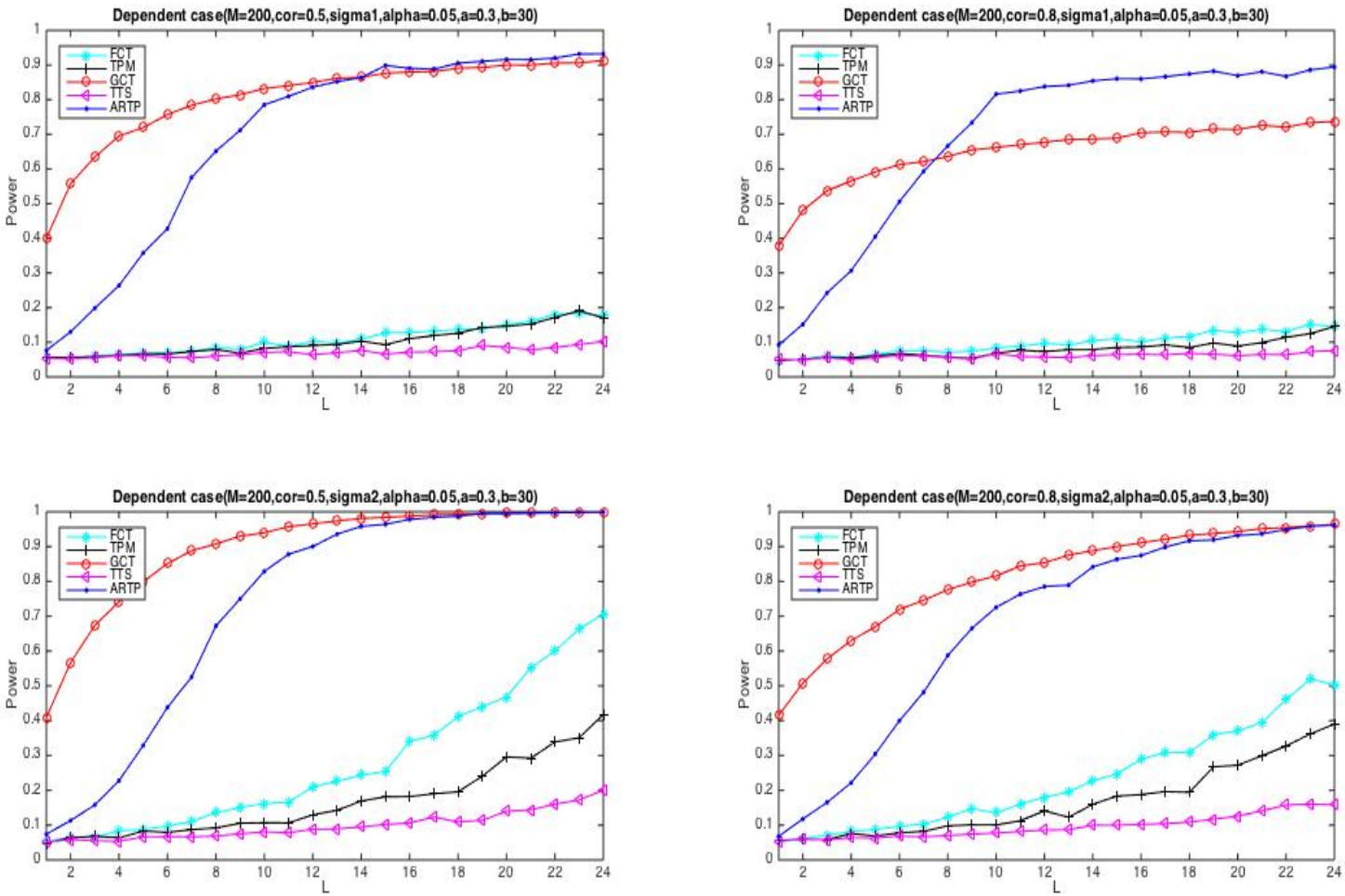
Power Comparison

Figure 4: Power comparison of Scenario 4



Power Comparison

Figure 5: Power comparison of Scenario 5



THANKS!